Group 16

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Battery State-of-Charge Estimation: From Data-Driven Models to Physics-Guided and Probabilistic Learning

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1 Introduction

Battery State of Charge (SOC) estimation is one of the most fundamental tasks in a battery management system (BMS). It plays a central role in ensuring the safe, efficient, and reliable operation of rechargeable battery systems used in electric vehicles, portable electronics, and large-scale energy storage applications. In practice, SOC indicates the amount of charge remaining in a battery relative to its available capacity, and it is closely related to several important functionalities, including driving-range prediction, charging management, power scheduling, and overcharge or over-discharge protection. Since the operating condition of a battery directly affects both system performance and safety, accurate SOC estimation has become a key requirement in modern battery-based technologies [1, 6, 7].

Unlike voltage, current, or temperature, SOC is not a directly measurable physical quantity. It cannot be obtained from a single sensor reading and instead must be inferred from observable signals combined with a suitable estimation model. This makes SOC estimation inherently challenging. The relationship between measurable variables and the true internal state of the battery is highly nonlinear, time-dependent, and influenced by many interacting factors. For example, the terminal voltage of a battery is affected not only by SOC, but also by load current, internal resistance, transient polarization effects, and ambient or internal temperature. Therefore, estimating SOC is not a simple regression problem, but rather a dynamic hidden-state estimation problem [2, 3].

Traditionally, model-based methods have been widely used for battery SOC estimation. These methods include ampere-hour integration (also known as coulomb counting), open-circuit-voltage-based lookup approaches, equivalent circuit models (ECMs), and filtering-based methods such as the Kalman filter and its variants. The advantage of these approaches lies in their physical interpretability and relatively low computational complexity. In engineering practice, they have served as effective solutions for many battery management systems. However, traditional methods often rely on simplified physical assumptions, parameter calibration, or fixed operating conditions. As battery behavior becomes more complicated under varying loads, dynamic temperature conditions, and realistic driving profiles, these assumptions may no longer be sufficient to achieve high estimation accuracy [1, 2, 3, 4].

In recent years, data-driven methods based on machine learning and deep learning have also attracted growing attention. Neural networks, recurrent models, and other sequence-learning architectures can learn nonlinear mappings from measured battery data to SOC without requiring an explicit hand-crafted physical model. These approaches are flexible and often achieve strong predictive performance when sufficient training data are available. Nevertheless, purely data-driven models also face important limitations. They may require large amounts of high-quality labeled data, suffer from poor generalization across battery chemistries or operating conditions, and produce predictions that are dif-

ficult to interpret physically. In safety-critical applications such as electric vehicles, these limitations are particularly important, because a prediction that is numerically accurate on average may still be unreliable or physically inconsistent in unseen situations [6, 7].

For this reason, current research is increasingly moving toward hybrid approaches that combine physical knowledge with modern learning-based techniques. Rather than viewing physics-based modelling and machine learning as competing alternatives, recent work attempts to integrate them in complementary ways. Differentiable models make it possible to embed battery dynamics into trainable computational frameworks, allowing model parameters to be optimized through gradient-based learning. Physics-guided scientific machine learning (SciML) incorporates known physical laws or constraints into neural architectures, helping improve generalization and physical plausibility. Probabilistic modelling further extends these approaches by providing uncertainty-aware predictions, which are important for robust decision-making in practical battery systems [5, 7, 10].

In this report, we focus specifically on the SOC estimation problem rather than joint SOC/SOH estimation. While battery State of Health (SOH) is also an important topic, SOC itself already presents a sufficiently rich and challenging problem due to its hidden-state nature, nonlinear system dynamics, and sensitivity to operating conditions. By concentrating on SOC estimation, we can more clearly examine how physics-based understanding, differentiable learning, and probabilistic reasoning contribute to improved battery state estimation. The goal is not only to achieve high prediction accuracy, but also to develop methods that are physically meaningful, robust under real-world conditions, and reliable enough for deployment in practical battery management systems.

2 Battery SOC Estimation Problem

The battery SOC estimation problem can be understood as the task of inferring an internal electrochemical state from external measurements. A battery is a complex dynamical system, and many of its most important internal states cannot be observed directly. Among these, SOC is one of the most critical because it describes how much usable charge remains in the battery at a given moment. In general terms, SOC is defined as the ratio between the remaining charge and the nominal or available capacity of the battery [1]:

$$SOC = \frac{Q_{\text{remaining}}}{Q_{\text{nominal}}} \times 100\% \tag{1}$$

This definition is intuitive, but it does not immediately solve the estimation problem, because neither the exact remaining charge nor the effective capacity is directly known during normal battery operation. Instead, they must be estimated indirectly.

One of the most basic methods for SOC estimation is coulomb counting. In this method, SOC is updated by integrating the battery current over time [1]:

$$SOC_{t+1} = SOC_t - \frac{\eta \Delta t}{Q} I_t \tag{2}$$

where η denotes coulombic efficiency, Δt is the sampling interval, Q is the battery capacity, and I_t is the measured current. Although this formulation is straightforward and computationally efficient, it has several weaknesses. First, it depends heavily on the accuracy of the initial SOC value. If the initial value is incorrect, the estimation error may persist for a long time. Second, any bias or noise in current measurement accumulates over time, causing drift. Third, the capacity Q used in the model may not remain constant in practice, especially when battery condition, temperature, or operating regime changes. As a result, coulomb counting alone is rarely sufficient for robust real-world SOC estimation [1, 4].

Another major difficulty lies in the relationship between SOC and terminal voltage. In principle, open-circuit voltage (OCV) is correlated with SOC, which suggests that voltage measurements can help estimate SOC. However, in practical operation, the measured terminal voltage is rarely equal to the OCV because the battery is usually under load. The observed voltage is influenced by multiple factors, including ohmic drop, polarization voltage, transient electrochemical effects, and hysteresis. This means that the same SOC can correspond to different voltage values depending on whether the battery is charging or discharging, how large the current is, and how recently the load has changed. Conversely, the same measured voltage may correspond to different SOC levels under different operating conditions. Therefore, mapping voltage directly to SOC is not straightforward [2, 3].

Temperature introduces another layer of complexity. Battery performance is highly temperature-dependent, and both electrochemical reactions and internal resistance are affected by thermal conditions. At low temperatures, internal resistance may increase significantly, changing the voltage response and reducing apparent available capacity. At high temperatures, reaction dynamics and degradation mechanisms may behave differently. Because of this, a model calibrated at one temperature may not perform well at another. Even if the estimation target is only SOC, the influence of temperature cannot be ignored, since the relationship among current, voltage, and SOC is no longer fixed across environments [4, 6].

In addition to nonlinearity and temperature dependence, battery SOC estimation is also complicated by time-varying system characteristics. Real batteries do not behave like ideal static devices. Their parameters change over time due to usage history, partial charge-discharge cycles, rest periods, and internal aging-related effects. Although this report does not focus on SOH estimation, it is still important to recognize that capacity fade and resistance increase indirectly affect SOC estimation performance. If a model assumes constant parameters, its accuracy may degrade when deployed over longer periods or under different operating conditions. This makes SOC estimation not only a static

input-output mapping problem, but a time-evolving state estimation problem requiring adaptation and robustness [7].

Measurement noise and data quality issues further complicate the problem. In real applications, current, voltage, and temperature sensors all have limited precision. Data may be sampled at different rates, affected by communication latency, or contaminated by disturbances from the surrounding system. In electric vehicles, for example, load profiles can change rapidly due to acceleration, braking, road conditions, and auxiliary power demands. Under such transient conditions, the battery response becomes more dynamic, and estimation models must remain stable despite short-term fluctuations. Therefore, an effective SOC estimator should not only minimize average prediction error, but also maintain robustness in the presence of sensor noise and abrupt operating changes.

From a systems perspective, the SOC estimation problem has several essential requirements. First, the estimator must capture nonlinear battery dynamics rather than relying on overly simplified assumptions. Second, it should remain physically reasonable, for example by avoiding impossible SOC values or unrealistic temporal behavior. Third, it should generalize across varying load conditions, temperatures, and usage scenarios. Finally, for safety-critical applications, the estimator should ideally provide some measure of confidence or uncertainty, since an overconfident but wrong estimate may be more dangerous than a conservative one [10].

For these reasons, battery SOC estimation is best viewed as a structured dynamic inference problem at the intersection of physical modelling, sequence learning, and uncertainty quantification. A successful solution must balance data adaptability with physical consistency. This is exactly why hybrid approaches have become increasingly important in recent research. By combining battery physics, differentiable learning methods, and probabilistic modelling, it becomes possible to develop SOC estimation frameworks that are more accurate, more robust, and more trustworthy than purely traditional or purely black-box methods alone [5, 7, 10].

3 Methodology and Design

3.1 Pure Data-Driven Models

3.1.1 Characteristics and Potential Limitations of Pure Data-Driven Approaches

Pure data-driven approaches estimate battery SOC by learning a direct mapping from measured signals to the target state, without explicitly incorporating electrochemical principles or state evolution equations. In this work, the input at time step t is defined as

$$x_t = [V_t, I_t, T_t, C_t, I_t^{load}, V_t^{load}],$$

and the SOC estimate is written as

$$S\hat{O}C_t = f_\theta(x_t),$$

where $f_\theta(\cdot)$ denotes a nonlinear regression model such as Random Forest or Multi-Layer Perceptron (MLP).

This formulation is simple, flexible, and able to provide strong baseline performance. However, as discussed in previous reviews, pure data-driven SOC estimation may still face several methodological limitations [6, 7].

First, physical consistency is not explicitly enforced. Although the model can learn a useful nonlinear mapping from data, it does not directly encode electrochemical constraints such as charge conservation or internal battery dynamics.

Second, in this work the problem is formulated as a current-timestep regression task, where each sample is constructed only from present measurements. This design simplifies implementation, but it may weaken the ability of the model to capture temporal effects such as historical current influence, polarization, and dynamic state transitions.

Third, model performance may depend on the training data distribution. When operating conditions differ from those represented in the training set, such as temperature, load profile, aging level, or battery type, the learned mapping may generalize less reliably.

Finally, the interpretability of such models is limited compared with physics-based formulations. Even when predictive accuracy is acceptable, it is often difficult to explain errors in terms of underlying physical mechanisms.

Overall, pure data-driven approaches are useful and effective baseline methods for SOC estimation, but they may still be limited in physical consistency, temporal modeling, generalization, and interpretability. These considerations motivate the comparison with more physics-guided approaches in the following sections.

3.1.2 Model Implementation of Pure Data-Driven Approaches

Based on the formulation described above, two representative data-driven models were implemented in this work: a Random Forest regressor and a Multi-Layer Perceptron (MLP). Both models take the current-timestep feature vector x_t as input and output a point estimate of SOC.

Random Forest Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. In this work, the Random Forest model is used as a nonparametric baseline to capture nonlinear relationships between the input features and SOC.

The model consists of 300 decision trees, each trained on bootstrapped samples of the training data. No explicit depth constraint is imposed, allowing the trees to fully adapt to the data. During inference, the final prediction is obtained by averaging the outputs

of all trees. This ensemble strategy improves robustness and reduces variance compared to a single decision tree.

Multi-Layer Perceptron (MLP) To provide a stronger nonlinear baseline, a fully connected neural network was implemented using PyTorch. The network architecture consists of multiple hidden layers with ReLU activations:

$$\begin{aligned}\text{Input} &\to \text{Linear}(d, 64) \to \text{ReLU}\\ &\to \text{Linear}(64, 64) \to \text{ReLU}\\ &\to \text{Linear}(64, 32) \to \text{ReLU}\\ &\to \text{Linear}(32, 1) \text{ (SOC output)}.\end{aligned}$$

where d is the number of input features.

The model is trained using the Adam optimizer with a learning rate of 10^{-3} and weight decay of 10^{-5}. Mean squared error (MSE) is used as the training objective. An early stopping strategy is applied based on validation loss, with a patience of 30 epochs, and the best-performing model on the validation set is restored.

Data Preprocessing The dataset is divided by cycle using a $6:2:2$ ratio for training, validation, and testing, ensuring that temporal leakage between cycles is avoided.

3.2 Physics-Based Model

This method focuses on battery state-of-charge (SOC) estimation based on a physically interpretable dynamic model. Rather than learning a black-box mapping directly from sensor measurements to SOC, the method updates SOC recursively using Coulomb counting [1]:

$$S\hat{O}C_{k+1} = S\hat{O}C_k + \frac{I_k \Delta t_k}{Q_n},$$

where I_k denotes the measured current, Δt_k is the time interval between two consecutive samples, and Q_n is the battery capacity used for normalization. Because the discharge current is negative in the dataset, the SOC decreases naturally during the discharge process. To ensure physical consistency, the estimated SOC is constrained to the interval $[0, 1]$.

In the implementation, the SOC trajectory is reconstructed for each cycle using the initial SOC, measured current, sampling interval, and capacity information. The reconstructed SOC is then compared with the reference SOC provided in the dataset. The estimation performance is evaluated using root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2).

To support future extensions toward dynamic battery modeling, a simple voltage-response model based on the OCV-SOC relationship and a first-order Thevenin equivalent

circuit was also considered. However, since the primary objective of this study is SOC estimation, voltage modeling is treated only as an auxiliary component rather than the main focus of analysis.

Overall, this method preserves physical interpretability while remaining computationally simple and efficient, making it suitable for structured battery SOC estimation.

3.3 Physics-guided Scientific Machine Learning (SciML)

Physics-guided Scientific Machine Learning (SciML) integrates physical knowledge into machine learning models so that the predictions are not only data-accurate but also physically consistent. For battery SOC estimation, this idea is particularly important because SOC is governed by charge conservation and current integration, rather than being an arbitrary regression target [5, 7].

In our implementation, a neural network is used to predict SOC from measured battery signals:

$$S\hat{O}C_t = f_\theta(x_t),$$

where the input feature vector is

$$x_t = [V_t, I_t, T_t, C_t, I_t^{load}, V_t^{load}].$$

The neural network is implemented as a multi-layer perceptron with several hidden layers and ReLU activations, allowing it to learn complex nonlinear relationships between the measured variables and SOC.

To incorporate physical knowledge, an additional physics-based loss is introduced. For each battery cycle, the predicted SOC trajectory is sorted by time, and the difference between consecutive predicted SOC values is computed as

$$\Delta S\hat{O}C_k = S\hat{O}C_{k+1} - S\hat{O}C_k.$$

According to the Coulomb counting principle, the physically expected SOC variation should satisfy

$$\Delta SOC_k^{phys} = -\frac{I_k^{avg}\Delta t_k}{C_k^{avg} \cdot 3600},$$

where I_k^{avg} and C_k^{avg} denote the average current and average capacity between two adjacent time steps, and Δt_k is the corresponding time interval.

Based on this idea, the physics loss is defined as

$$\mathcal{L}_{phys} = \frac{1}{N-1}\sum_{k=1}^{N-1}\left[(S\hat{O}C_{k+1} - S\hat{O}C_k) - \Delta SOC_k^{phys}\right]^2.$$

The standard supervised data loss is defined as the mean squared error between predicted

and true SOC:

$$\mathcal{L}_{data} = \frac{1}{N}\sum_{t=1}^{N}(S\hat{O}C_t - SOC_t)^2.$$

The total training objective combines the two terms:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_{phys}\mathcal{L}_{phys},$$

where λ_{phys} controls the trade-off between data fitting and physical consistency.

Compared with purely data-driven neural networks, this SciML framework encourages the predicted SOC trajectory to obey a physically meaningful evolution law, which improves interpretability and can enhance robustness under noisy or limited data conditions. At the same time, the neural network retains the flexibility to capture nonlinear relationships that are difficult to model using simple equivalent-circuit equations alone.

However, the current implementation still uses a simplified physical constraint based only on SOC difference consistency. More complex battery dynamics, such as polarization, hysteresis, and electrochemical diffusion, are not explicitly modeled. In addition, the physics loss is computed outside the end-to-end automatic differentiation graph, so the framework should be viewed as a practical physics-guided learning approach rather than a fully differentiable physics-constrained model.

Overall, SciML provides a useful compromise between black-box learning and rigid physics-based modeling, making it a promising direction for reliable battery SOC estimation.

3.4 Probabilistic Modelling for Uncertainty in SOC Estimation

Probabilistic modelling aims to estimate not only the battery SOC itself, but also the uncertainty associated with that estimate. In practical battery management systems, a single point prediction is often insufficient, because the reliability of the estimate may vary under different operating conditions, noise levels, temperatures, and battery aging states. Therefore, instead of predicting only

$$S\hat{O}C_t,$$

probabilistic methods aim to model a conditional distribution

$$p(SOC_t \mid x_t),$$

where x_t denotes the observed battery measurements [10].

A common probabilistic formulation assumes that the SOC follows a Gaussian distribution conditioned on the input:

$$SOC_t \mid x_t \sim \mathcal{N}(\mu_t, \sigma_t^2),$$

where μ_t is the predictive mean and σ_t^2 is the predictive variance. In this case, the model outputs both the estimated SOC and its uncertainty:

$$(\mu_t, \sigma_t^2) = f_\theta(x_t).$$

Instead of minimizing only mean squared error, the model can be trained by maximizing the data likelihood, which is equivalent to minimizing the negative log-likelihood:

$$\mathcal{L}_{NLL} = \frac{1}{N} \sum_{t=1}^{N} \left[\frac{(SOC_t - \mu_t)^2}{2\sigma_t^2} + \frac{1}{2} \log \sigma_t^2 \right].$$

Several probabilistic methods are commonly used for battery state estimation. First, Bayesian filtering methods such as Kalman Filter (KF), Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Particle Filter (PF) treat SOC as a latent state and recursively update its posterior distribution using a state-space model [2, 3]:

$$SOC_{k+1} = f(SOC_k, I_k) + w_k,$$

$$V_k = h(SOC_k, I_k) + v_k,$$

where w_k and v_k represent process and measurement noise. These methods naturally provide both state estimates and uncertainty information.

Second, Bayesian neural networks model the network weights as random variables, so that the prediction itself becomes a distribution rather than a deterministic value. Third, practical approximations such as Monte Carlo Dropout estimate uncertainty by performing multiple stochastic forward passes during inference. Similarly, deep ensembles estimate uncertainty by training multiple independently initialized models and measuring the variation among their predictions.

An important advantage of probabilistic modelling is that it can distinguish between different sources of uncertainty. Aleatoric uncertainty reflects measurement noise and operating variability, while epistemic uncertainty reflects the model's lack of knowledge under unseen or poorly represented conditions. In battery SOC estimation, this distinction is useful because uncertainty often increases when the battery operates outside the training distribution or when the measurements are noisy [10].

Compared with deterministic models, probabilistic approaches provide confidence intervals and are better suited for risk-aware decision-making, anomaly detection, and safety-critical battery management. For example, a prediction interval such as

$$[\mu_t - 1.96\sigma_t,\ \mu_t + 1.96\sigma_t]$$

can help assess whether the estimated SOC is sufficiently reliable for downstream control.

However, probabilistic modelling also has limitations. These methods are usually

more computationally expensive, more difficult to train, and may require careful calibration to ensure that the predicted uncertainty truly reflects the actual prediction error. In addition, real-time deployment in embedded battery management systems may be constrained by computational resources.

Overall, probabilistic modelling provides an essential extension to deterministic battery state estimation by quantifying confidence in the prediction. This makes it a valuable direction for developing more robust and trustworthy SOC estimation systems [10].

4 Dataset Description

In this work, experiments are conducted on the *NASA Ames Prognostics Center of Excellence (PCoE) Li-ion Battery Aging Dataset* [8], using the subset denoted as **batch1** in our processed data organization. The original dataset was collected from a custom-built battery prognostics testbed at NASA Ames and is widely used for battery health monitoring, state estimation, and prognostics research. It contains run-to-failure measurements of lithium-ion cells under repeated aging experiments, where each battery undergoes charge, discharge, and electrochemical impedance spectroscopy (EIS) operations. During cycling, key variables such as terminal voltage, current, temperature, time, and capacity-related information are recorded, which makes the dataset suitable for modeling battery degradation behavior and evaluating data-driven estimation methods. In this study, the **batch1** subset is used as the experimental data source to analyze battery performance.

5 Results Analysis of Approaches

This section presents a quantitative and qualitative analysis of the results obtained from the different SOC prediction approaches.

Across all approaches, model performance was evaluated using a consistent set of criteria to ensure a fair and systematic comparison. Specifically, four quantitative metrics were reported in the tables: RMSE, MAE, R^2, and inference time. RMSE measures the overall magnitude of prediction error and places greater emphasis on larger deviations. MAE reflects the average absolute difference between the predicted and true SOC values. The coefficient of determination, R^2, indicates how well the model explains the variance in the SOC labels, with values closer to 1 suggesting a better fit. Inference time refers to the total wall-clock time required for prediction on the full test set, and therefore reflects computational efficiency during deployment. In general, better model performance is characterized by lower RMSE, lower MAE, lower inference time, and higher R^2.

In addition to the tabulated metrics, qualitative performance was examined through plots of predicted SOC versus ground-truth SOC for the first test cycle of each battery dataset, which provide a more intuitive view of each model's tracking accuracy and its ability to follow the temporal SOC variation throughout the discharge process.

5.1 Results Analysis of Pure Data-Driven Approaches

This section evaluates two pure data-driven baseline models for SOC prediction, namely RandomForest and MLP.

As reported in Tables 1 and 2, MLP consistently outperformed RandomForest across all four battery datasets. On average, RandomForest achieved an RMSE of 0.0617, an MAE of 0.0413, and an R^2 of 0.9584, whereas MLP reduced the average RMSE to 0.0376 and the average MAE to 0.0298, while increasing the average R^2 to 0.9844. This indicates that the neural baseline provided a noticeably more accurate mapping from measured variables to SOC than the tree-based baseline under the same pure data-driven setting. The improvement was especially clear on B0005 and B0006, where the reduction in prediction error was substantial, while B0018 remained the easiest dataset for both methods. One possible explanation is that B0018 contained fewer cycles than the other batteries, which may have reduced inter-cycle variability and made its SOC patterns easier for both models to learn. In addition, in terms of inference time, MLP achieved a lower computational cost on all four datasets during the inference stage.

Table 1: Test performance of the RandomForest model on the four battery datasets.

Battery	RMSE	MAE	R^2	Inference Time (s)
B0005	0.0873	0.0569	0.9271	0.061600
B0006	0.0772	0.0434	0.9452	0.061809
B0007	0.0544	0.0434	0.9690	0.077749
B0018	0.0280	0.0213	0.9922	0.077858
Average	0.0617	0.0413	0.9584	0.069754

Table 2: Test performance of the MLP model on the four battery datasets.

Battery	RMSE	MAE	R^2	Inference Time (s)
B0005	0.0327	0.0232	0.9898	0.000980
B0006	0.0575	0.0476	0.9696	0.000879
B0007	0.0417	0.0349	0.9818	0.001030
B0018	0.0185	0.0136	0.9966	0.001241
Average	0.0376	0.0298	0.9844	0.001032

The same pattern is evident in the first-test-cycle plots shown in Fig. 1 and Fig. 2. RandomForest was able to capture the overall decreasing SOC trend during discharge, indicating that it learned the basic mapping between the measured variables and battery state. However, its predictions showed a noticeable underestimation over much of the cycle. Moreover, near the end of discharge, irregular behaviors such as rebound or abrupt deviation were observed in B0005 and B0007, after the true SOC had already approached zero. This suggests that RandomForest was less robust in the low-SOC region and less reliable in modeling the discharge tail. A plausible explanation is that, as a tree-based ensemble, RandomForest performs piecewise averaging in the feature space and does not

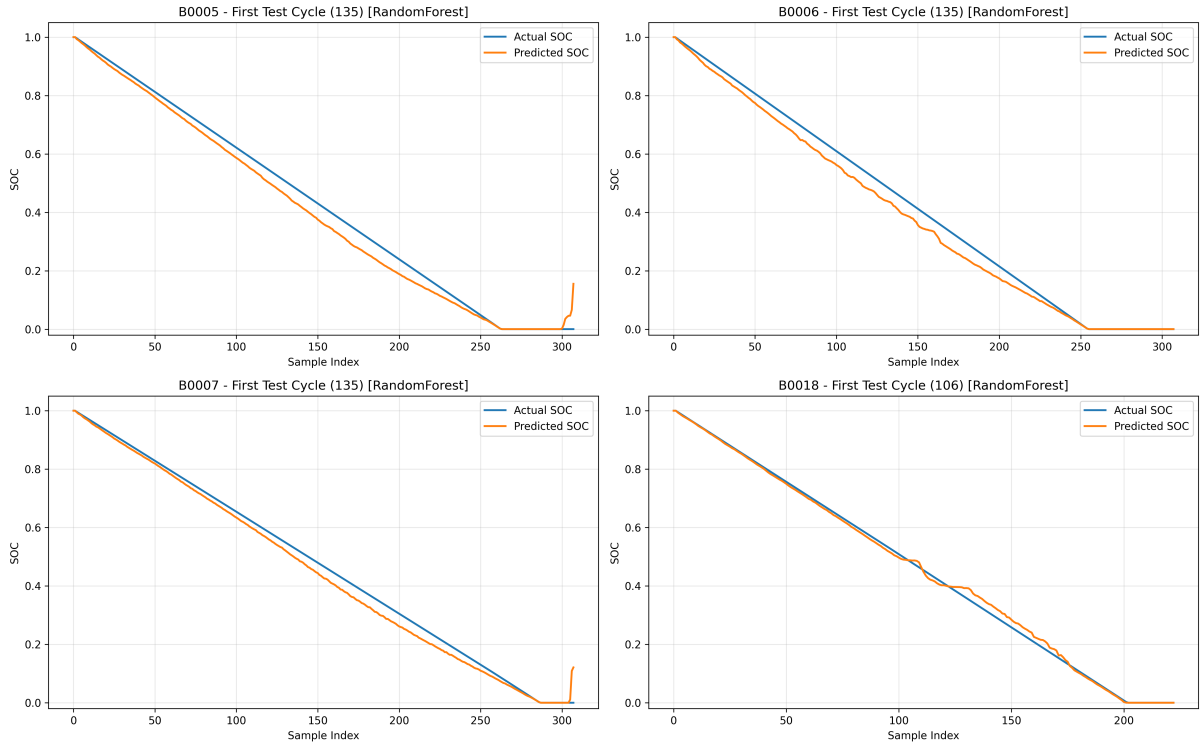

Figure 1: Predicted and ground-truth SOC trajectories for the first complete test cycle of each battery dataset, using the RandomForest model.

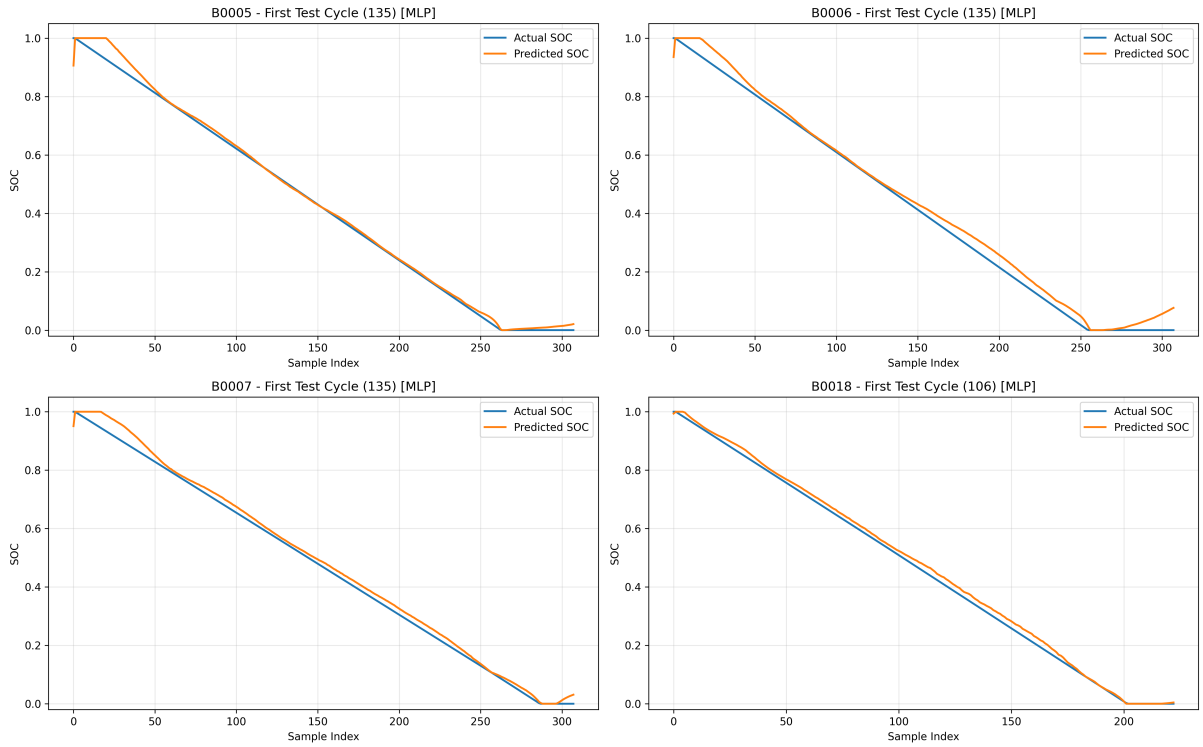

Figure 2: Predicted and ground-truth SOC trajectories for the first complete test cycle of each battery dataset, using the MLP model.

explicitly enforce temporal consistency or monotonicity. As a result, when the discharge process approaches the zero-SOC boundary, the model may produce small local artifacts instead of maintaining a physically consistent decay trend.

In comparison, MLP produced prediction curves that aligned more closely with the ground-truth SOC across the full first test cycle. For all four batteries, the predicted trajectories were smoother and showed smaller deviation from the reference profile throughout discharge. This advantage was especially apparent in the middle and late stages, where the model tracked the monotonic SOC decay more accurately and with reduced bias. However, minor overestimation remained at the beginning of some cases. This may be because the initial high-SOC region is more difficult to distinguish from instantaneous measurements alone, so the network tended to generate slightly elevated predictions at the start of discharge. Even so, unlike RandomForest, MLP did not exhibit the same rebound behavior near the end of discharge, which indicates better stability at the low-SOC boundary. The prediction on B0018 was particularly accurate, with the estimated curve nearly overlapping the ground truth over most of the cycle.

5.2 Results Analysis of the Physics-Based Model

The physics-based battery estimator achieves moderate SOC estimation performance across the four battery datasets. Averaged over all batteries, the model obtains a mean SOC RMSE of 0.1343, a mean SOC MAE of 0.1177, and a mean SOC R^2 of 0.7763. Among the four datasets, B0007 shows the best performance, with SOC RMSE = 0.1090, SOC MAE = 0.0953, and SOC R^2 = 0.8455, while B0006 is the most challenging case, with SOC RMSE = 0.1498 and SOC R^2 = 0.7148. Overall, the results indicate that the model can capture the general SOC trend, but its absolute estimation accuracy remains limited.

Table 3: Test performance of the Physics-based model on the four battery datasets.

Battery	RMSE	MAE	R^2	Inference Time (s)
B0005	0.1379	0.1208	0.7698	0.121595
B0006	0.1498	0.1318	0.7148	0.146566
B0007	0.1090	0.0953	0.8455	0.117741
B0018	0.1407	0.1230	0.7750	0.090406
Average	0.1343	0.1177	0.7763	0.119077

Figure 3 shows the predicted and ground-truth SOC trajectories for the first complete test cycle of each battery dataset. For all four batteries, the physics-based model captures the overall monotonic decreasing trend of SOC during discharge. However, the predicted SOC does not closely overlap with the ground-truth curve. Instead, it remains consistently higher than the true SOC over most of the cycle, and the gap gradually increases as discharge proceeds. This discrepancy is particularly visible near the end of

each cycle, where the true SOC approaches zero while the predicted SOC still stays at a noticeable positive level.

These observations suggest that the model can reproduce the coarse discharge trend on a simple first test cycle, but it still exhibits a systematic overestimation error. This bias is mainly caused by accumulated integration error and model mismatch in the assumed battery capacity and OCV relationship. Since the selected first-cycle trajectories are relatively smooth and nearly linear, they represent an easier case for Coulomb-counting-based state propagation. Therefore, the averaged quantitative metrics across all test cycles provide a more reliable assessment of the model's overall performance. In other words, although the model captures the general shape of SOC evolution, its accuracy remains limited when evaluated more comprehensively across different cycles and operating conditions.

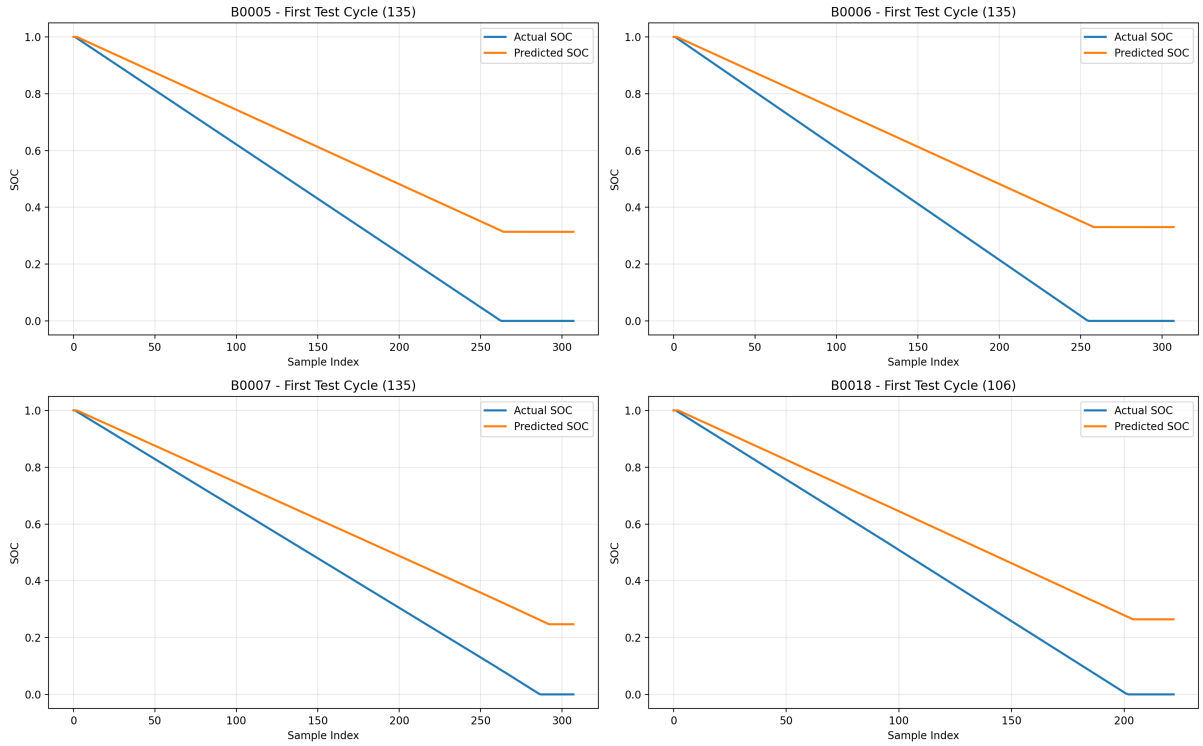

Figure 3: Predicted and ground-truth SOC trajectories for the first complete test cycle of each battery dataset, using the Physics-based model.

In terms of efficiency, the average SOC calculation time is approximately 0.119 s per dataset. However, this runtime is not directly comparable to the feed-forward inference time of the pure data-driven baselines, since the physics-based model involves an end-to-end sequential computation process, including state propagation, OCV-related calculations, parameter identification, and voltage simulation. Therefore, the reported timing should be interpreted as a file-level computational cost rather than a per-sample inference time.

Compared with the pure data-driven baselines, the physics-based model shows lower

predictive accuracy under the current experimental setting. Its average SOC MAE and RMSE are higher than those of both Random Forest and MLP, while its R^2 is clearly lower. In particular, the MLP baseline achieves substantially smaller errors and better goodness-of-fit, indicating that the data-driven models can make more effective use of the available measurements in this task. Overall, these results suggest that although the physics-based model provides stronger physical interpretability, its current first-order equivalent-circuit structure is not yet flexible enough to match the predictive accuracy of the stronger learning-based baselines.

5.3 Results Analysis of the SciML Model

Table 4 summarizes the final test performance of the SciML model on the four battery datasets in terms of RMSE, MAE, R^2, and total inference time on the test set. Averaged over the four batteries, the model achieves a test RMSE of 0.0322, a test MAE of 0.0242, and a test R^2 of 0.9880. In addition, the average total inference time for the whole test set is only 0.000766 seconds, showing that the model remains very lightweight during inference.

Table 4: Test performance of the SciML model on the four battery datasets.

Battery	RMSE	MAE	R^2	Inference Time (s)
B0005	0.0291	0.0219	0.9919	0.000879
B0006	0.0576	0.0441	0.9695	0.000832
B0007	0.0236	0.0162	0.9941	0.000791
B0018	0.0186	0.0145	0.9966	0.000560
Average	0.0322	0.0242	0.9880	0.000766

Among the four batteries, B0018 shows the best overall performance, with test RMSE = 0.0186, test MAE = 0.0145, and test R^2 = 0.9966, indicating an almost perfect agreement between predicted and actual SOC values. B0007 also performs very well, with low estimation error and a high coefficient of determination. B0005 remains highly accurate as well, achieving stable SOC prediction throughout the test set. In contrast, B0006 is still the most challenging dataset, with the largest test RMSE and MAE, together with the lowest R^2 among the four batteries. Nevertheless, even for B0006, the model still achieves an R^2 above 0.96, which indicates that the overall prediction quality remains strong.

To further examine the prediction quality, Fig. 4 plots the first complete cycle from the test set for each battery. In all four cases, the predicted SOC closely follows the true discharge trajectory, which is consistent with the high R^2 values reported in Table 4. This demonstrates that the model is not only effective in terms of aggregate numerical metrics, but also capable of capturing the time-varying SOC evolution within an individual discharge cycle.

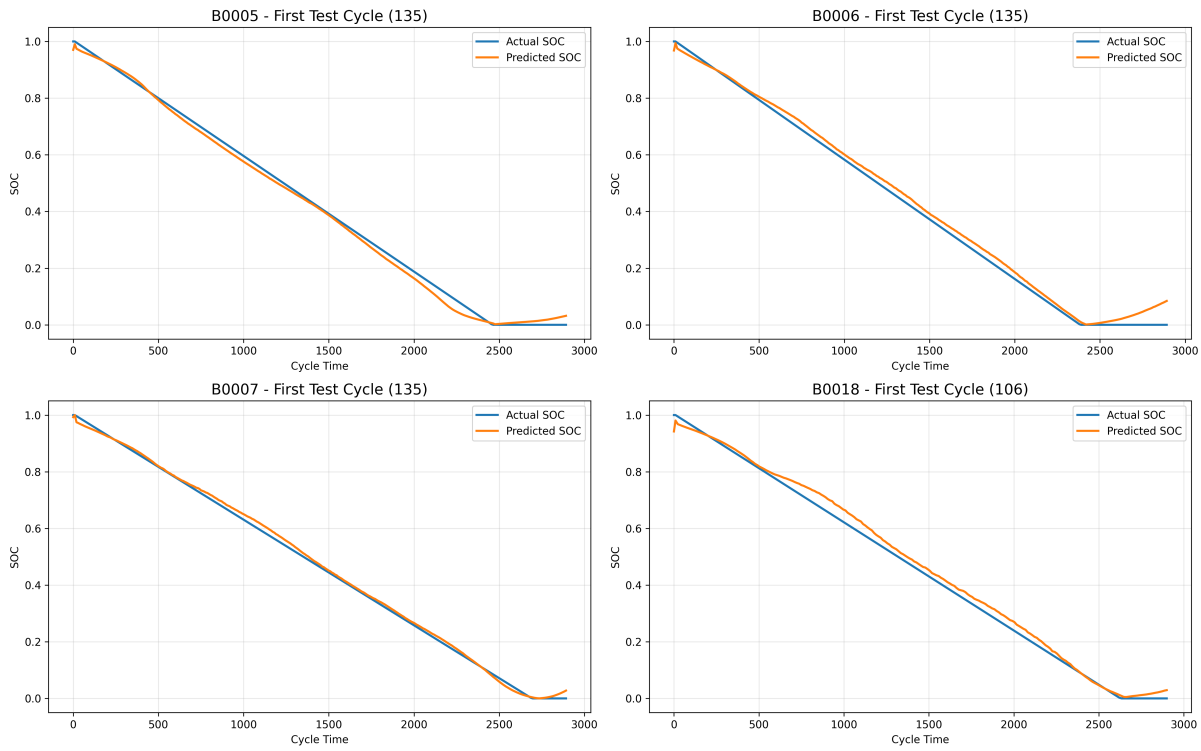

Figure 4: Predicted SOC versus ground-truth SOC for the first complete test cycle of each battery dataset, using the SciML model.

More specifically, for B0005, B0006, and B0007, the model successfully reproduces the overall monotonic discharge trend, while small deviations can still be observed in the middle-to-late stage of discharge. These deviations become more visible when SOC approaches zero, where slight overestimation appears near the end of the cycle. Such behavior is most noticeable for B0006, which is consistent with its relatively larger numerical error in Table 4. In contrast, the prediction for B0018 almost completely overlaps with the ground truth throughout the cycle, confirming that this battery is the easiest for the model to fit. Therefore, the figure-based analysis supports the metric-based conclusion that the SciML model provides highly accurate SOC prediction across different batteries, while still leaving limited residual error in the more difficult cases.

Compared with the differentiable/model-based approach, the SciML model shows a clear improvement in predictive performance. Its average MAE and RMSE are substantially reduced, while the average R^2 increases to 0.9880. This suggests that the learning-based formulation adopted in the SciML module is much more effective than relying only on a simplified first-order equivalent-circuit model.

Compared with the pure data-driven baselines, the SciML model clearly outperforms Random Forest and remains competitive with MLP. Relative to Random Forest, it achieves lower average MAE and RMSE together with a higher R^2. Relative to MLP, the difference is relatively small, indicating that the proposed model is able to deliver strong predictive accuracy while maintaining a structured modeling framework. Overall,

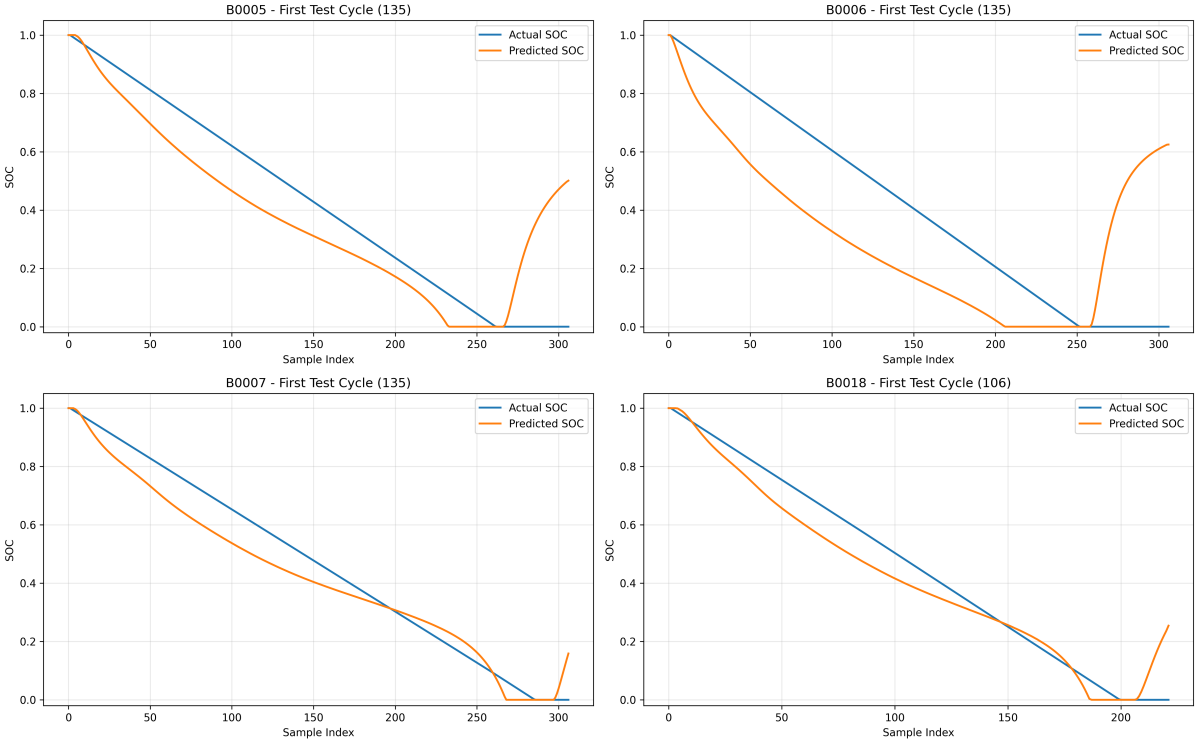

Figure 5: First test-cycle SOC prediction results of the probabilistic model on the four battery datasets, using the probabilistic model.

the results show that this SciML model provides an effective and efficient solution for SOC estimation under the considered experimental setting.

5.4 Results Analysis of the Probabilistic Model

The probabilistic model shows the weakest point-estimation performance among all methods evaluated in this study. Averaged across the four battery datasets, it achieves an RMSE of 0.1740, an MAE of 0.1426, and an R^2 of 0.6405, with an average inference time of 0.7113 s. Among the four batteries, B0007 gives the best result, with RMSE = 0.0948, MAE = 0.0737, and R^2 = 0.9059, while B0006 is the most difficult case, with RMSE = 0.3166, MAE = 0.2706, and R^2 = 0.0786.

As shown in Fig. 5, the probabilistic model is able to capture the general downward trend of SOC during the first test cycle on all four battery datasets. However, its trajectory quality is clearly weaker than that of the other models. For all four batteries, the predicted SOC is systematically lower than the ground truth over a large portion of the discharge process, indicating a consistent underestimation bias. In addition, a physically implausible rebound appears near the end of discharge in all cases, where the predicted SOC rises again after approaching zero. This behavior suggests that the current probabilistic model lacks sufficient physical constraints to preserve a realistic discharge trajectory.

Table 5 further confirms that the point-estimation accuracy of the probabilistic model

Table 5: Test performance of the probabilistic model on the four battery datasets.

Battery	RMSE	MAE	R^2	Inference Time (s)
B0005	0.1829	0.1463	0.6804	0.8069
B0006	0.3166	0.2706	0.0786	0.7963
B0007	0.0948	0.0737	0.9059	0.8041
B0018	0.1018	0.0797	0.8969	0.4377
Average	0.1740	0.1426	0.6405	0.7113

remains limited under the current setup. Compared with the physical-based model, the probabilistic model does not improve deterministic prediction accuracy, as its RMSE and MAE are higher and its R^2 is lower. Compared with Random Forest, MLP, and especially SciML, the gap is even larger, with noticeably higher errors and weaker goodness-of-fit. In terms of computation, the probabilistic model also introduces a larger inference cost than the deterministic baselines, which is consistent with the additional burden of uncertainty-aware prediction, such as estimating distribution parameters or performing repeated stochastic inference.

Overall, these results indicate that the current probabilistic implementation does not yet provide competitive point-estimation performance for SOC prediction. Nevertheless, this conclusion should be interpreted with caution. The main value of probabilistic modelling lies not only in minimizing deterministic metrics such as RMSE and MAE, but also in providing meaningful uncertainty quantification. Therefore, without additional evaluation of uncertainty calibration, prediction interval coverage, and confidence reliability, the full benefit of the probabilistic model is not yet fully reflected in the current comparison.

6 Conclusion

This report investigated several representative approaches for battery SOC estimation, including pure data-driven models, physics-based methods, physics-guided scientific machine learning (SciML), and probabilistic modelling. The study aimed to compare these methods from multiple perspectives, including prediction accuracy, inference efficiency, physical consistency, and modeling interpretability.

The experimental results show that pure data-driven methods provide strong baseline performance. Among them, the MLP achieved great overall point-estimation accuracy, with the lower average MAE and RMSE and higher average R^2 across the four battery datasets. It also demonstrated very fast inference, making it an effective solution when predictive accuracy and deployment efficiency are the main priorities. Random Forest also performed reasonably well, but it was consistently weaker than MLP in both error metrics and inference speed.

The physics-based approach offered the strongest physical interpretability, since it

explicitly modeled SOC evolution, OCV-SOC relationship, and battery voltage dynamics through a first-order equivalent-circuit structure. However, its prediction accuracy was clearly limited under the current setting. Although it could follow the general discharge trend and produced visually plausible trajectories on some cycles, its overall RMSE and MAE remained significantly higher than those of the learning-based approaches. This suggests that a simplified first-order circuit model is not flexible enough to fully capture the nonlinear behavior of the batteries considered in this study.

The SciML approach achieved the best balance between learning flexibility and physical consistency. By combining neural-network-based nonlinear regression with a physics-guided SOC evolution constraint, it significantly improved performance over the physics-based method and outperformed Random Forest. Although SciML achieved slightly better point-estimation accuracy than the MLP, the performance gap was relatively small. Therefore, SciML emerged as the most balanced approach in this work, since it preserved much of the predictive strength of data-driven learning while incorporating physically meaningful structure [5, 7].

The probabilistic model, in contrast, showed the weakest point-estimation performance among the evaluated methods. Its prediction curves captured the overall downward SOC trend, but systematic underestimation and physically implausible rebound near the end of discharge were frequently observed. In addition, its RMSE and MAE were substantially higher than those of MLP and SciML. However, this result should be interpreted with caution: the primary value of probabilistic modelling lies in uncertainty quantification rather than only deterministic point prediction. Since the current evaluation mainly focused on RMSE, MAE, and R^2, the potential benefits of uncertainty-aware estimation were not fully reflected [10].

Overall, the comparison reveals a clear trade-off among the four categories of methods. SciML achieve the strongest raw predictive accuracy and fastest inference. Pure data-driven models MLP achieved the second-best overall performance. Pure physics-based methods provide higher interpretability but lower flexibility. Probabilistic methods are promising for uncertainty-aware battery management, but their advantages require more comprehensive uncertainty evaluation metrics to be properly assessed.

In conclusion, for the current SOC estimation task, MLP is the strongest choice when only prediction accuracy is considered and when a large amount of high-quality data is available, while SciML is the most promising direction when both accuracy and physical consistency are important. Future work can further improve these approaches by introducing stronger physical constraints, better uncertainty calibration, and more advanced hybrid frameworks that combine data-driven learning, differentiable dynamics, and probabilistic reasoning in a unified battery state estimation system.

7 Contributions

The contributions of each group member are summarized as follows.

- **Liu Yingying (A0318511M)**: responsible for the code implementation and documentation writing of the *Pure Data-Driven Approaches* section.
- **Zhan Zhenrong (A0319143J)**: responsible for the model implementation and documentation writing of the *Physics-Based Model* section.
- **Zhang Chi (A0318254E)**: responsible for the implementation and documentation writing of the *Scientific Machine Learning* section.
- **Cai Yuansheng (A0329930Y)**: responsible for the code implementation and documentation writing of the *Probabilistic Model* section.
- **Yu Wendi (A0319038A)**: responsible for writing the *Introduction*, *Background*, and the comparative summary of model performance.

8 Declaration

During the preparation of this blog, we used GPT-5 to assist in understanding relevant concepts, organizing the structure of the article, and improving the clarity of language expression. All group members reviewed, revised, and take full responsibility for the final submitted content.

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